

Air University



SSDD (Lab)

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Lab Task # 05

Lab Task

- You have to make a state diagram for the given scenario, keeping in view all the system requirements.
- For your help, the important components that should be present in your state diagram are also given below.

Title: "Secure Access Control System State Chart Diagram Case Study"

Introduction:

In this case study, we will explore the design and implementation of a state chart diagram for a secure access control system. The system is intended to manage access to a corporate office building, ensuring that only authorized personnel can enter restricted areas. Security is paramount, and the state chart diagram will play a crucial role in visualizing the system's behavior and ensuring that it meets security requirements.

Background:

ApexDefender is a company that operates a high-security office building where sensitive information is stored, and critical operations take place. They have decided to implement a modern access control system to replace the outdated manual system. The new system will use electronic keycards and biometric authentication methods to grant access to employees based on their roles and permissions.

System Requirements:

1. Only authorized personnel should be allowed access to restricted areas.
2. Employees must use their keycards or biometric data for authentication.
3. The system should log all access attempts and generate alerts for suspicious activities.
4. In case of a security breach or unauthorized access, the system should initiate an immediate response, such as sounding alarms and notifying security personnel.
5. The system should be able to manage access for various areas within the building, each with its own set of permissions.
6. Access permissions should be easily configurable by authorized administrators.
7. The system should be scalable to accommodate future expansion and evolving security needs.

Components of State Chart Diagram:

The state chart diagram will represent the different states and transitions within the access control system. Here are some key components of the diagram:

1. Initial State:

- Represents the starting point of the system when it's powered on and in a standby mode.

2. Authentication States:

- Idle: The system is waiting for an employee to present their keycard or biometric data.
- Authenticating: The system is in the process of verifying the employee's identity.
- Authenticated: The employee's identity has been verified successfully, and access is granted.

3. Access Control States:

- Access Granted: The authorized personnel are allowed entry.

- **Access Denied:** Unauthorized personnel are denied entry, and an alert is generated.
- **Alarm Activated:** If a security breach is detected, alarms are activated, and security personnel are notified.

4. Administrator Configuration States:

- **Configuration Mode:** Authorized administrators can enter this state to modify access permissions.
- **Saving Configuration:** The system is saving the changes made by administrators.
- **Configuration Saved:** Confirmation that the new configurations have been successfully saved.

5. Emergency Response States:

- **Emergency Mode:** Triggered in response to a security breach, initiating a predefined emergency response protocol.
- **Lockdown:** A state of high security where all access is restricted except for authorized security personnel.

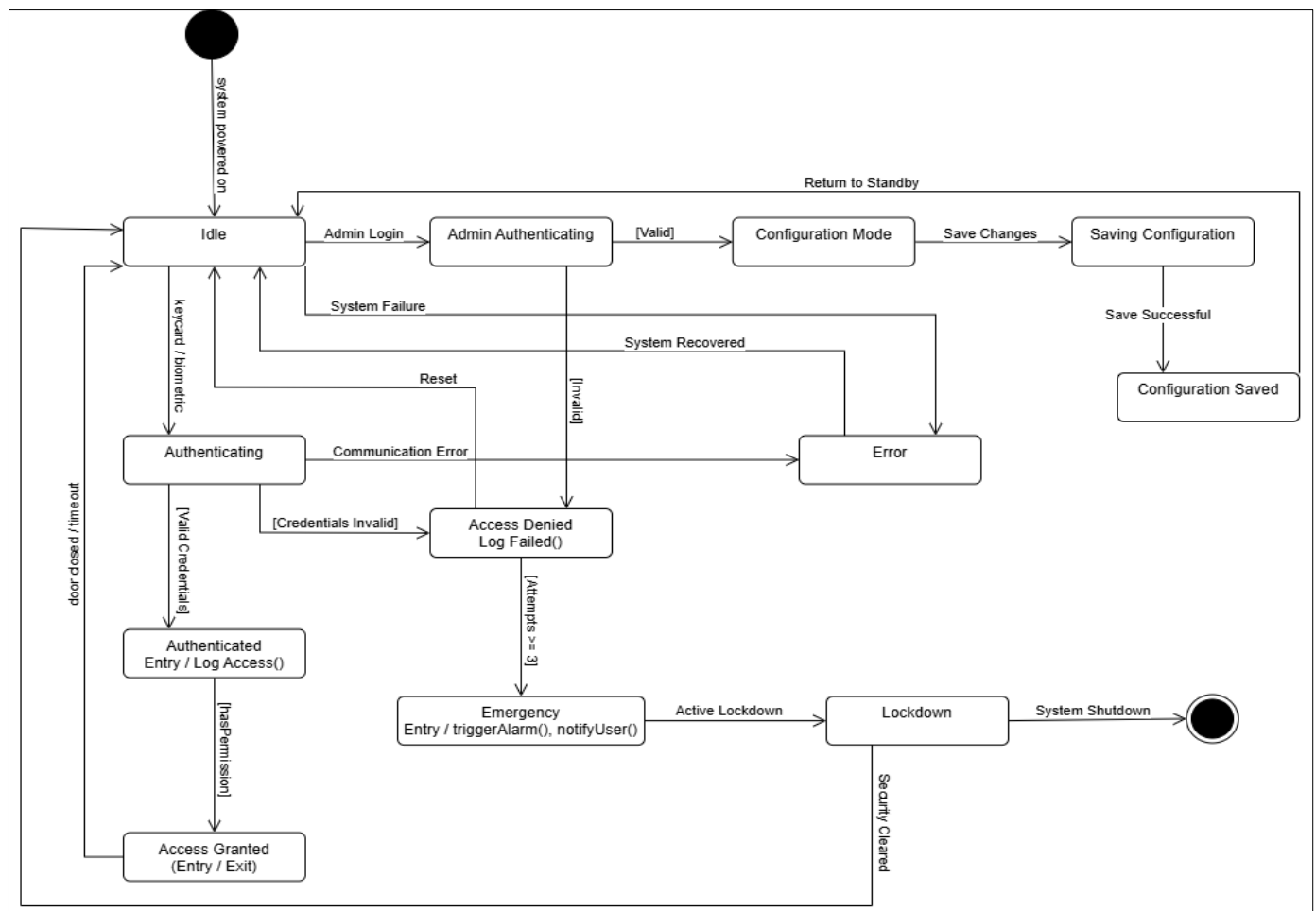
6. Error States:

- **Error:** Represents various error conditions that may occur, such as system malfunction or communication issues.

Conclusion:

The state chart diagram for the secure access control system will provide a visual representation of the system's behavior and interactions. This diagram will serve as a valuable tool for system designers, developers, and security experts to ensure that the access control system meets the stringent security requirements of ApexDefender Corporation. It will also aid in troubleshooting and maintaining the system in the future.

Solution



Explanation:

This diagram shows how a secure access control system works for a high-security corporate building. The system ensures that only authorized personnel can enter restricted areas while managing authentication, configuration, error handling, and emergency response.

1. Initial and Idle State

When the system is turned on, it enters the Initial State, then transitions to the Idle State. In the Idle state, the system waits for:

- Employee authentication (using a keycard or biometric input)
- Administrator login
- System reset or recovery events

This state represents the system's standby mode.

2. Employee Authentication Flow

When an employee uses a keycard or biometric data, the system goes into the Authenticating state.

- If the credentials are correct, the system moves to the Authenticated state, allowing access.
- If the credentials are incorrect, the system transitions to the Access Denied state, recording the failed attempt.

From the authenticated state, the system checks area permissions:

- If permission is granted, access is allowed (Entry/Exit).
- After the door closes or after a timeout, the system returns to Idle.

This ensures that only those with the right permissions can enter restricted areas.

3. Suspicious Activity & Emergency Handling

If there are three or more invalid access attempts, the system enters the Emergency state. This triggers an alarm and alerts security personnel, leading to Lockdown mode.

- In Lockdown, all access is restricted, allowing only authorized security personnel to respond.
- Once security resolves the situation, the system returns to Idle.

Alternatively, in the event of a security breach, the system may go directly to a shutdown state.

4. Administrator Configuration Flow

In the Idle state, an administrator can log in. The system checks the administrator's credentials.

- If the credentials are valid, the system enters Configuration Mode, allowing changes to be saved.
- After saving successfully, the system logs the Configuration Saved message and goes back to Idle.

This allows authorized administrators to easily manage and update access permissions.\

5. Error Handling

If the system encounters failures or communication errors, it goes into the Error state. Once these issues are resolved, the system returns to Idle.

This feature improves the system's reliability in case of problems.

Conclusion

The state chart diagram highlights:

- Control over authentication and authorization
- Logging of access attempts
- Procedures for emergencies and lockdowns in response to suspicious activity
- Management of administrator configurations
- Detection and recovery from errors

Overall, the diagram demonstrates the security, reliability, and access control needed in a high-security corporate environment.

THE END
